Ellipses

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Standard form**

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:

a the vertices $(\pm 5, 0)$

b the foci $(\pm 4, 0)$

c the endpoints of the minor axis $(0, \pm 3)$

d the eccentricity $e = \frac{4}{5}$

(a) $(\pm 5, 0)$; (b) $(\pm 4, 0)$; (c) $(0, \pm 3)$; (d) $e = \frac{4}{5}$

2 **[Foundational]** • **Target: Standard form**

Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.

Major axis vertical with $a = 5$, $c = 3$, so $b^2 = 25 - 9 = 16$: $\frac{x^2}{16} + \frac{y^2}{25} = 1$.

3 **[Foundational]** • **Target: Standard form**

Consider the ellipse $9x^2 + 4y^2 = 36$.

a Write the equation in standard form. $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

b State the lengths of the major and minor axes. Major axis $2a = 6$ (vertical); minor axis $2b = 4$.

c Find the coordinates of the foci. $c^2 = 9 - 4 = 5$, so foci $(0, \pm \sqrt{5})$.

(a) $\frac{x^2}{4} + \frac{y^2}{9} = 1$.; (b) Major axis $2a = 6$ (vertical); minor axis $2b = 4$.; (c) $c^2 = 9 - 4 = 5$, so foci $(0, \pm \sqrt{5})$.

4 **[Foundational]** • **Target: Applications**

A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?

$c = \sqrt{400 - 144} = \sqrt{256} = 16$, so the foci are $2c = 32.0$ m apart.

5 **[Foundational]** • **Target: Applications**

An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.

$a = 6$: $\frac{9}{36} + \frac{12}{b^2} = 1$ gives $\frac{12}{b^2} = \frac{3}{4}$, so $b^2 = 16$: $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

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6 [Foundational] • Target: Ellipses centered away from the origin

For the ellipse $\frac{(x-2)^2}{16}+\frac{(y+1)^2}{9}=1$, find:

a the center $(2,-1)$ **b** the vertices $(2\pm4,-1)=(6,-1),(-2,-1)$ **c** the foci $c=\sqrt{16-9}=\sqrt{7}$: $(2\pm\sqrt{7},-1)$

(a) $(2,-1)$; (b) $(2\pm4,-1)=(6,-1),(-2,-1)$; (c) $c=\sqrt{16-9}=\sqrt{7}$: $(2\pm\sqrt{7},-1)$

7 [Foundational] • Target: Ellipses centered away from the origin

Complete the square to write $4x^2+y^2-8x+6y+9=0$ in standard form, and state the center.

$4(x^2-2x)+(y^2+6y)=-9$: $4(x-1)^2-4+(y+3)^2-9=-9$, so $4(x-1)^2+(y+3)^2=4$, i.e. $(x-1)^2+\frac{(y+3)^2}{4}=1$. Center: $(1,-3)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Eccentricity and orbital applications

Compute the eccentricity of each ellipse, and state which is more circular (closer to $e=0$):

a $\frac{x^2}{100}+\frac{y^2}{36}=1$ $c=\sqrt{100-36}=8$: $e=\frac{8}{10}=0.8$ **b** $\frac{x^2}{25}+\frac{y^2}{24}=1$ $c=\sqrt{25-24}=1$: $e=\frac{1}{5}=0.2$ — this one is more circular

(a) $c=\sqrt{100-36}=8$: $e=\frac{8}{10}=0.8$; (b) $c=\sqrt{25-24}=1$: $e=\frac{1}{5}=0.2$ — this one is more circular

9 [Intermediate] • Target: Eccentricity and orbital applications

A planet's elliptical orbit has semi-major axis $a=150$ million km and eccentricity $e=0.02$. Find the distance from the center to a focus, c, and hence the closest approach distance to the sun ($a-c$), to two decimal places (in millions of km).

$c=ea=0.02(150)=3$ million km. Closest approach $=a-c=150-3=147.00$ million km.

10 [Intermediate] • Target: Graphing an ellipse

The upper half of the ellipse $\frac{x^2}{16}+\frac{y^2}{4}=1$ is shown (as $y=2\sqrt{1-x^2/16}$). Use it to estimate the y-value at $x=2$, then verify algebraically.

From the graph, $y\approx1.7$ at $x=2$. Algebraically: $y=2\sqrt{1-4/16}=2\sqrt{0.75}\approx1.73$ ✓.

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11 [Intermediate] • Target: The ellipse's defining property

State the geometric (sum-of-distances) definition of an ellipse, and use it to verify that the point $(0, 3)$ lies on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with foci $(\pm\sqrt{7}, 0)$.

An ellipse is the set of points where the sum of distances to the two foci is constant ($= 2a$). At $(0, 3)$: distance to each focus $= \sqrt{7 + 9} = 4$, so the sum is $8 = 2a = 2(4)$ ✓, confirming the point lies on the ellipse.

12 [Intermediate] • Target: Writing equations from a description

Write the standard-form equation of an ellipse centered at $(3, 2)$ with a horizontal major axis of length 10 and minor axis of length 6.

$a = 5$, $b = 3$: $\frac{(x-3)^2}{25} + \frac{(y-2)^2}{9} = 1$.

13 [Intermediate] • Target: Writing equations from a description

Write the standard-form equation of an ellipse with vertical major axis, center $(-1, 4)$, $a = 7$, and $b = 2$.

$\frac{(x+1)^2}{4} + \frac{(y-4)^2}{49} = 1$.

14 [Intermediate] • Target: Systems with ellipses

Find the points of intersection of $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the line $y = x$.

Substitute $y = x$: $\frac{x^2}{9} + \frac{x^2}{4} = 1 \Rightarrow \frac{4x^2 + 9x^2}{36} = 1 \Rightarrow 13x^2 = 36 \Rightarrow x = \pm\frac{6}{\sqrt{13}}$. Points: $\left(\pm\frac{6}{\sqrt{13}}, \pm\frac{6}{\sqrt{13}}\right)$ (same sign pairs).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 9.02 Geometric Sequences and Series

State the formula for the area of an ellipse with semi-axes a and b, and use it to find the area of $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

Area $= \pi ab$. Here $a = 5$, $b = 4$: Area $= 20\pi \approx 62.83$ square units.

16 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 9.03 The Binomial Theorem

An elliptical track for a running field has a semi-major axis of 60 m and semi-minor axis of 40 m. Find the exact area enclosed by the track, and give a decimal approximation to the nearest square metre.

Area $= \pi(60)(40) = 2400\pi \approx 7539.8$ m^2.

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17 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 1.01 Functions and Function Notation

Two ellipses both have eccentricity $e = 0.6$: one has $a = 10$, the other has $a = 20$. Find b for each, and explain why they are the same shape but different sizes (similar ellipses).

For $a = 10$: $c = ea = 6$, $b^2 = 100 - 36 = 64 \Rightarrow b = 8$. For $a = 20$: $c = 12$, $b^2 = 400 - 144 = 256 \Rightarrow b = 16$. Since $\frac{b}{a} = \frac{8}{10} = \frac{16}{20} = 0.8$ is the same ratio for both, the ellipses are geometrically similar (same shape, scaled by a factor of 2).

18 [Challenging] • Target: Eccentricity and orbital applications • Cross-Lesson: 1.02 Domain and Range

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$. $\cot(2\theta) = \frac{A - C}{B} = \frac{5 - 8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$. Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1 + \cos(2\theta)}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}},$.

19 [Challenging] • Target: Graphing an ellipse • Cross-Lesson: 1.03 Average Rate of Change

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer k. Inductive Step: We need to show that $a_{k+1} = (k + 1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k + 1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers n. Closed-form expression for S_N: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

20 [Challenging] • Target: The ellipse's defining property • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi + 2\pi k}{4}\right) + i\sin\left(\frac{\pi + 2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right],\ k \in \{0, 1, 2, 3\}$. Step 3: Evaluate for each k:$k = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Horizontal and Vertical Standard Forms

Find the center, vertices, and co-vertices of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. **Center:** $(0, 0)$**, Vertices:** $(\pm 5, 0)$**, Co-vertices:** $(0, \pm 3)$ ✓

b. Center: $(0, 0)$, Vertices: $(0, \pm 5)$, Co-vertices: $(\pm 3, 0)$

c. Center: $(5, 3)$, Vertices: $(\pm 5, 0)$

d. Center: $(0, 0)$, Vertices: $(\pm 25, 0)$

$a^2 = 25 \Rightarrow a = 5$ (major axis horizontal). $b^2 = 9 \Rightarrow b = 3$. Vertices at $(\pm 5, 0)$, co-vertices at $(0, \pm 3)$.

22 [Foundational] • Target: Horizontal and Vertical Standard Forms

Find the standard equation of the ellipse with center $(0, 0)$, vertices $(0, \pm 6)$, and co-vertices $(\pm 4, 0)$.

a. $\frac{x^2}{16} + \frac{y^2}{36} = 1$ ✓

b. $\frac{x^2}{36} + \frac{y^2}{16} = 1$

c. $\frac{x^2}{4} + \frac{y^2}{6} = 1$

d. $\frac{x^2}{20} + \frac{y^2}{36} = 1$

Major axis vertical: $a = 6 \Rightarrow a^2 = 36$ under y^2. Minor axis horizontal: $b = 4 \Rightarrow b^2 = 16$ under x^2.

23 [Foundational] • Target: Horizontal and Vertical Standard Forms

Find the length of the major and minor axes for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{4} = 1$.

a. **Major axis** $= 8$**, Minor axis** $= 4$ ✓

b. Major axis $= 4$, Minor axis $= 2$

c. Major axis $= 16$, Minor axis $= 4$

d. Major axis $= 8$, Minor axis $= 2$

$a^2 = 16 \Rightarrow a = 4$, major axis length $2a = 8$. $b^2 = 4 \Rightarrow b = 2$, minor axis length $2b = 4$.

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24 [Foundational] • Target: Foci and the Fundamental Relationship

What is the fundamental relationship between semi-major axis a, semi-minor axis b, and focal distance c in an ellipse?

a. $c^2 = a^2 - b^2$ ✓

b. $c^2 = a^2 + b^2$

c. $a^2 = b^2 - c^2$

d. $c = a - b$

In an ellipse, the foci lie on the major axis inside the vertices, governed by the Pythagorean relation $c^2 = a^2 - b^2$.

25 [Foundational] • Target: Foci and the Fundamental Relationship

Find the coordinates of the foci of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. $(\pm 4, 0)$ ✓

b. $(0, \pm 4)$

c. $(\pm\sqrt{34}, 0)$

d. $(\pm 16, 0)$

$c^2 = a^2 - b^2 = 25 - 9 = 16 \Rightarrow c = 4$. Major axis is horizontal, so foci are $(\pm 4, 0)$.

26 [Foundational] • Target: Foci and the Fundamental Relationship

Find the foci of $\frac{x^2}{16} + \frac{y^2}{25} = 1$.

a. $(0, \pm 3)$ ✓

b. $(\pm 3, 0)$

c. $(0, \pm\sqrt{41})$

d. $(\pm 9, 0)$

Major axis is vertical ($a^2 = 25$). $c^2 = 25 - 16 = 9 \Rightarrow c = 3$. Foci are $(0, \pm 3)$.

27 [Foundational] • Target: Eccentricity

How is the eccentricity e of an ellipse defined, and what is its range?

a. $e = \frac{c}{a}$, **where** $0 \le e < 1$ ✓

b. $e = \frac{a}{c}$, where $e > 1$

c. $e = \frac{b}{a}$, where $e < 0$

d. $e = c^2/a^2$

Eccentricity is the ratio of focal distance to semi-major axis: $e = c/a$. For any ellipse, $0 \le e < 1$ (with $e = 0$ being a circle).

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28 [Foundational] • Target: Eccentricity

Find the eccentricity of the ellipse $\frac{x^2}{100} + \frac{y^2}{64} = 1$.

a. 0.6 ✓

b. 0.8

c. 0.36

d. 0.64

$a = 10, b = 8 \Rightarrow c = \sqrt{100 - 64} = \sqrt{36} = 6$. Eccentricity $e = c/a = 6/10 = 0.6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Eccentricity

What happens to the shape of an ellipse as its eccentricity approaches 0?

a. **It becomes nearly circular** ✓

b. It becomes very elongated / needle-like

c. It approaches a parabola

d. It opens outward

When $e \to 0$, $c \to 0$, meaning the two foci merge into the center, producing a circle ($a = b$).

30 [Intermediate] • Target: Completing the Square and Whispering Galleries

Convert $4x^2 + 9y^2 - 16x + 54y + 61 = 0$ to standard form.

a. $\frac{(x-2)^2}{9} + \frac{(y+3)^2}{4} = 1$ ✓

b. $\frac{(x-2)^2}{4} + \frac{(y+3)^2}{9} = 1$

c. $\frac{(x+2)^2}{9} + \frac{(y-3)^2}{4} = 1$

d. $(x-2)^2 + (y+3)^2 = 36$

$4(x^2 - 4x + 4) + 9(y^2 + 6y + 9) = -61 + 16 + 81 = 36$. Dividing by 36: $\frac{(x-2)^2}{9} + \frac{(y+3)^2}{4} = 1$.

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31 [Intermediate] • Target: Completing the Square and Whispering Galleries

A whispering gallery is an elliptical room 100 ft long and 60 ft wide. How far from the center are the two whispering spots (foci)?

a. 40 ft ✓

b. 50 ft

c. 30 ft

d. 20 ft

$2a = 100 \Rightarrow a = 50$. $2b = 60 \Rightarrow b = 30$. $c = \sqrt{50^2 - 30^2} = \sqrt{2500 - 900} = \sqrt{1600} = 40$ ft from the center.

32 [Intermediate] • Target: Completing the Square and Whispering Galleries

Why can a whisper at one focus of an elliptical gallery be heard clearly at the other focus?

a. **Sound waves reflecting off the elliptical wall all converge simultaneously at the other focus** ✓

b. Sound waves travel faster along ellipses

c. The walls absorb echo

d. The ceiling acts as a resonator

By the reflective property of the ellipse, any ray emitted from one focus reflects off the perimeter directly toward the opposite focus with equal path length.

33 [Intermediate] • Target: Eccentricity

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

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34 [Intermediate] • Target: Completing the Square and Whispering Galleries

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. **Consistent and dependent (infinitely many solutions) ✓**

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

35 [Intermediate] • Target: Standard form

Two lines have slopes $m_1 = 2$, $m_2 = -3$ and y-intercepts $b_1 = -2$, $b_2 = 0$. Classify the system.

a. **Consistent and independent (one solution) ✓**

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

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36 [Intermediate] • Target: Applications

Two lines have slopes $m_1 = -4$, $m_2 = 0$ and y-intercepts $b_1 = 1$, $b_2 = 2$. Classify the system.

a. **Consistent and independent (one solution) ✓**

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

37 [Intermediate] • Target: Ellipses centered away from the origin

Two lines have slopes $m_1 = 1$, $m_2 = 1$ and y-intercepts $b_1 = -1$, $b_2 = -1$. Classify the system.

a. Consistent and independent (one solution)

b. **Consistent and dependent (infinitely many solutions) ✓**

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Eccentricity and orbital applications • Cross-Lesson: 1.02 Domain and Range

Two lines have slopes $m_1 = 0$, $m_2 = -2$ and y-intercepts $b_1 = 1$, $b_2 = 3$. Classify the system.

a. Consistent and independent (one solution) ✓

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

39 [Challenging] • Target: Graphing an ellipse • Cross-Lesson: 1.03 Average Rate of Change

Two lines have slopes $m_1 = -1$, $m_2 = -4$ and y-intercepts $b_1 = 3$, $b_2 = 5$. Classify the system.

a. Consistent and independent (one solution) ✓

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

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40 [Challenging] • Target: The ellipse's defining property • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Two lines have slopes $m_1 = 4$, $m_2 = -3$ and y-intercepts $b_1 = 0$, $b_2 = 2$. Classify the system.

a. Consistent and independent (one solution) ✓

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

41 [Challenging] • Target: Writing equations from a description • Cross-Lesson: 2.02 Radian Measure and Arc Length

Two lines have slopes $m_1 = 0$, $m_2 = 4$ and y-intercepts $b_1 = -2$, $b_2 = -3$. Classify the system.

a. Consistent and independent (one solution) ✓

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

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42 [Challenging] • Target: Systems with ellipses • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Two lines have slopes $m_1 = -4$, $m_2 = -3$ and y-intercepts $b_1 = 2$, $b_2 = 1$. Classify the system.

a. Consistent and independent (one solution) ✓

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

43 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Two lines have slopes $m_1 = 0$, $m_2 = -3$ and y-intercepts $b_1 = -2$, $b_2 = -1$. Classify the system.

a. Consistent and independent (one solution) ✓

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

44 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4° ✓

In the right triangle, opposite $= 12$, hypotenuse $= 13$, so the angle is $\sin^{-1}\left(\frac{12}{13}\right) \approx 67.4^\circ$.

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45 [Challenging] • Target: Horizontal and Vertical Standard Forms • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A 10 m ladder leans against a vertical wall and reaches 7 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 44.4° ✓

b. 35.0°

c. 45.6°

d. 49.4°

In the right triangle, opposite $= 7$, hypotenuse $= 10$, so the angle is $\sin^{-1}\left(\frac{7}{10}\right) \approx 44.4^\circ$.

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